Nebojsa Jovanovic

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Professional Summary

Mechanical Engineer with a Bachelor of Applied Science in Mechanical Engineering and consulting engineering experience at Hatch across multidisciplinary industrial, infrastructure, and energy projects. Experienced in package engineering, mechanical design support, tunnel fire ventilation, commissioning, technical documentation, and coordination among engineering, construction, and project stakeholders. Contributed to large-scale project execution by managing technical deliverables, supporting systems integration, resolving documentation gaps, and improving engineering workflows through automation. Combines mechanical engineering fundamentals with software development, industrial control systems, and applied AI to support technical problem-solving and digital engineering initiatives. Hatch Engineering Entrance Scholarship recipient with a collaborative approach to engineering delivery, continuous improvement, and multidisciplinary project execution.

Technical Skills

- **Mechanical Engineering:** Mechanical Engineering Principles, Package Engineering, Mechanical Systems, Tunnel Fire Ventilation, Systems Integration, Engineering Design Support
- **Project Delivery:** Engineering Deliverables, Project Coordination, Technical Documentation, Commissioning Support, Systems Handover, Schedule Coordination, Stakeholder Collaboration
- **Multidisciplinary Coordination:** Engineering and Construction Interfaces, Cross-Disciplinary Collaboration, Technical Requirements Interpretation, Information Management, Issue Resolution
- **Infrastructure and Industrial Projects:** Mineral Processing, Lithium and Copper Processing, Potash Processing, Energy Systems, Tunnel Infrastructure, Industrial Facilities
- **Engineering Documentation and Tools:** SmartPlant Foundation, Smart Completions, SharePoint, Azure DevOps
- **Digital Engineering and Automation:** Python, C++, C#, SQL, Workflow Automation, Data Processing, Systems Integration
- **Industrial Systems and Controls:** Industrial Automation, Control Systems, Real-Time Data Systems, System Monitoring, Performance Analysis
- **AI and Digital Innovation:** Machine Learning, Computer Vision, Edge Computing, AI Model Deployment, Engineering Workflow Automation
- **Core Competencies:** Analytical Problem-Solving, Technical Communication, Attention to Detail, Initiative, Collaboration, Continuous Improvement, Technical Documentation Quality

Professional Experience

Hatch Ltd. – *Mechanical Engineer / Package Engineer / Commissioning Engineer / Software Developer*

Toronto, ON | Jan 2020 – Jul 2026

- Contributed to multidisciplinary engineering and project delivery activities across mineral processing, lithium and copper processing, energy systems, and infrastructure projects.
- Performed package engineering activities for projects including Lilac, Li-Cycle, and Manjar Maju Refinery, supporting engineering coordination and the organization of technical information and deliverables.
- Contributed to tunnel fire ventilation design for the La Fontayne Tunnel and other projects, gaining experience in mechanical engineering applications for infrastructure and fire-safety systems.
- Supported commissioning and project execution for the large-scale Jansen potash project as a Handover Coordinator within the HBJV commissioning team.
- Coordinated technical documentation and system handover activities to support the transition of completed systems into commissioning and operational readiness.
- Prepared Handover Packages, Subsystem Test Packages, and Notices of Energization (NOEs) to support project execution, system readiness, and commissioning deliverables.
- Coordinated information across engineering, construction, commissioning, and project stakeholders to support delivery requirements and resolve documentation gaps.

- Used SmartPlant Foundation, Smart Completions, and SharePoint to organize technical documentation, maintain project information, and facilitate coordination between project participants.
- Developed automation for repetitive engineering and project documentation workflows, including automated generation of NOE documents, improving consistency and reducing manual administrative effort.
- Collaborated with multidisciplinary teams to identify workflow inefficiencies, investigate technical and documentation issues, and implement practical improvements.
- Contributed to VISTA, a real-time hydro energy plant operating and optimization system developed using C++ and C#, integrating software functionality with industrial control systems.
- Implemented data-ingestion and performance-analysis modules supporting industrial system monitoring and analysis.
- Maintained and modernized legacy software components, improving code maintainability and supporting continued system development.
- Worked within an Azure DevOps-driven development environment supporting software development, testing, and deployment.
- Developed and deployed computer vision capabilities for the Onaping Depth Mine, applying machine learning to industrial safety and operational-efficiency applications.
- Built edge computing systems for field-level monitoring and execution of machine learning models in industrial environments.
- Applied software development and automation techniques to improve internal technical processes, data handling, and engineering workflows.
- Worked across engineering and technology activities to support systems integration, technical problem-solving, and continuous improvement in project delivery.

SELECTED PROJECTS

Jansen Potash Project — Commissioning and Systems Handover

Hatch / HBJV

- Worked on-site as a Handover Coordinator within the HBJV commissioning team on a large-scale industrial project.
- Coordinated systems handover activities and technical deliverables required to support commissioning and operational readiness.
- Prepared Handover Packages, Subsystem Test Packages, and Notices of Energization.
- Coordinated documentation and information requirements across engineering, construction, commissioning, and project stakeholders.
- Used SmartPlant Foundation, Smart Completions, and SharePoint to manage project information and support execution activities.
- Automated repetitive documentation processes, including NOE generation, to improve workflow efficiency and consistency.
- Supported project delivery by identifying documentation gaps, organizing technical deliverables, and coordinating information required for system handover.

Package Engineering — Mineral Processing Projects

Hatch

- Contributed to package engineering activities for lithium and copper processing projects, including Lilac, Li-Cycle, and Manjar Maju Refinery.
- Supported coordination of technical information and engineering deliverables within multidisciplinary project teams.
- Worked across engineering interfaces to help organize project information and support technical execution requirements.
- Developed experience contributing to complex industrial engineering projects involving multiple disciplines and project stakeholders.

La Fontayne Tunnel — Fire Ventilation Design

Hatch

- Contributed to tunnel fire ventilation design activities.
- Gained exposure to mechanical ventilation engineering considerations for infrastructure and fire-safety applications.
- Worked within a multidisciplinary engineering environment involving technical requirements and coordination across project disciplines.

- Developed experience relevant to mechanical systems and infrastructure engineering.

VISTA — Energy Plant Optimization Tool

Hatch

- Contributed to a real-time hydro energy plant operating and optimization system developed using C++ and C#.
- Implemented data-ingestion and performance-analysis functionality to support industrial system monitoring.
- Worked with control logic and industrial automation system interfaces.
- Maintained and modernized legacy software components supporting energy system operations.
- Collaborated within a development team using Azure DevOps for software development, testing, and deployment.

Onaping Depth Mine — Computer Vision Platform

Hatch

- Developed computer vision capabilities using Python, Flask, PyTorch, and Angular for industrial mining applications.
- Implemented camera-based inspection and analysis functionality using machine learning models.
- Supported deployment of machine learning applications in industrial environments.
- Applied analytical problem-solving and software engineering to develop tools supporting operational efficiency and safety-related applications.

SELECTED DIGITAL ENGINEERING PROJECTS

VISP — Computer Vision Platform on Edge Devices

Personal Project

- Designed and developed an on-premises platform for deploying computer vision applications on edge devices.
- Implemented functionality for managing RTSP cameras and querying stored detections.
- Designed a modular architecture supporting higher-level applications such as alerts, analytics, and operational dashboards.
- Integrated machine learning, networking, databases, and web services into a unified deployment platform.

Custom C++ Framework

Personal Project

- Built a lightweight HTTP server and in-memory database engine in C++.
- Developed practical knowledge of software architecture, networking, data structures, and memory management.
- Applied structured problem-solving to the design and implementation of systems-level software.

Resume Automation

Personal Project

- Developed an automated workflow to convert a Markdown resume into a professionally formatted PDF.
- Applied scripting and document-generation techniques to improve consistency and reduce repetitive formatting work.

Education

University of Toronto — Bachelor of Applied Science in Mechanical Engineering

Minor in Robotics and Mechatronics | Toronto, ON | 2020

- Hatch Engineering Entrance Scholarship — Full Ride Scholarship Recipient.
- Developed foundational knowledge of mechanical engineering, systems integration, robotics, and mechatronics through undergraduate education.

Continuous Education

- **Harvard CS50:** Completed CS50x, CS50 Python, CS50 Web, and CS50 SQL.

- **edX:** Completed online coursework in Data Structures and Algorithms.
- **Boot.dev:** Completed programming coursework involving guided and independent projects.
- **Nand2Tetris:** Completed coursework covering computer architecture, machine language, assembly, compiler development, and operating-system fundamentals.

Interests

Building Services Engineering, Mechanical Systems, Infrastructure Design, Fire Ventilation, Sustainable Engineering, Energy Systems, Multidisciplinary Project Delivery, Commissioning, Digital Engineering, Applied AI, Engineering Workflow Automation, Industrial Automation, and Systems Integration.